TransY-Net: Learning Fully Transformer Networks for Change Detection of Remote Sensing Images

Tianyu Yan, Zifu Wan, Pingping Zhang, Gong Cheng, *Member, IEEE*, and Huchuan Lu, *Senior Member, IEEE*

***Abstract*— In the remote sensing field, change detection (CD) aims to identify and localize the changed regions from dual-phase images over the same places. Recently, it has achieved great progress with the advances of deep learning. However, current methods generally deliver incomplete CD regions and irregular CD boundaries due to the limited representation ability of the extracted visual features. To relieve these issues, in this work we propose a novel transformer-based learning framework named TransY-Net for remote sensing image CD, which improves the feature extraction from a global view and combines multilevel visual features in a pyramid manner. More specifically, the proposed framework first uses the advantages of transformers in long-range dependency modeling. It can help learn more discriminative global-level features and obtain complete CD regions. Then, we introduce a novel pyramid structure to aggregate multilevel visual features from transformers for feature enhancement. The pyramid structure grafted with a progressive attention module (PAM) can improve the feature representation ability with additional interdependencies through spatial and channel attentions. Finally, to better train the whole framework, we use the deeply supervised learning with multiple boundary-aware loss functions. Extensive experiments demonstrate that our proposed method achieves a new state-of-the-art performance on four optical and two SAR image CD benchmarks. The source code is released at https://github.com/Drchip61/TransYNet.**

***Index Terms*— Change detection (CD), deep learning, progressive attention, remote sensing image, vision transformer.**

I. Introduction

CHANGE detection (CD) plays an important role in the field of remote sensing. It aims to detect the key change regions in dual-phase remote sensing images captured at different times but over the same scene area. In fact, remote sensing image CD has been used in many real-world applications, such as land-use planning, urban expansion management, geological disaster monitoring, and ecological environment protection. However, because the change regions can be any shapes in complex scenarios, there are still many challenges for CD. In addition, remote sensing image CD by handcrafted methods is time-consuming and labor-intensive, and thus, there is a great need for fully automatic and highly efficient CD.

In recent years, deep learning has been widely used in remote sensing image processing due to its powerful feature representation capabilities and has shown great potential in CD. With convolutional neural networks (CNNs) [1] and Transformers [2], [3], many CD methods extract more discriminative features and have demonstrated good performances. However, previous CD methods still have the following shortcomings: 1) With the resolution improvement of remote sensing images, rich semantic information contained in high-resolution images is not fully used. As a result, current CD methods are unable to distinguish pseudochanges such as shadow, vegetation, and sunshine in sensitive areas. 2) Boundary information in complex remote sensing images is often missing. In previous methods, the extracted changed areas often have regional holes and their boundaries can be very irregular, resulting in a poor visual effect [4]. 3) The temporal information contained in dual-phase remote sensing images is not fully used, which is also one of the reasons for the low performance of current CD methods.

To tackle the above issues, in this work we propose a novel transformer-based learning framework named TransY-Net for remote sensing image CD, which improves the feature extraction from a global view and combines multilevel visual features in a pyramid manner. More specifically, the proposed framework has a Y-shape structure whose input is a dual-phase remote sensing image pair. We first use the advantages of transformers in long-range dependency modeling to learn more discriminative global-level features. Then, to highlight the change regions, the summation features and difference features are generated by directly comparing the temporal features of dual-phase remote sensing images. Thus, one can obtain complete CD regions. To improve the boundary perception ability, we further introduce a pyramid structure to aggregate multilevel visual features from transformers. The pyramid structure grafted with a progressive attention module (PAM) can improve the feature representation ability with additional interdependencies through spatial and channel

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Tianyu Yan, Zifu Wan, and Pingping Zhang are with the School of Artificial Intelligence, Dalian University of Technology, Dalian 116024, China (e-mail: 2981431354@mail.dlut.edu.cn; 2537998622@qq.com; zhpp@dlut.edu.cn).

Gong Cheng is with the School of Automation, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: gcheng@nwpu.edu.cn).

Huchuan Lu is with the School of Information and Communication Engineering, Dalian University of Technology, Dalian 116024, China (e-mail: lhchuan@dlut.edu.cn).

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attentions. Finally, to better train the framework, we use the deeply supervised learning with multiple boundary-aware loss functions. Extensive experiments show that our method achieves a new state-of-the-art performance on four optical and two SAR image CD benchmarks.

The main contributions are summarized as follows.

1) We propose a transformer-based learning framework (i.e., TransY-Net) for remote sensing image CD, which can improve the feature extraction from a global view and combine multilevel features in a pyramid manner.
2) We propose a novel pyramid structure grafted with a PAM to further improve the feature representation ability with additional interdependencies through spatial and channel attentions.
3) We introduce the deeply supervised learning with multiple boundary-aware loss functions, to address the irregular boundary problem in CD.
4) Extensive experiments on four optical and two SAR image CD benchmarks show that our framework attains better performances than most state-of-the-art methods.

We note that this work is an extension of its previous conference version [5] with some key improvements as follows: 1) we propose a more powerful PAM with joint spatial and channel attentions; 2) we enhance the multilevel visual features with multiscale pooling; 3) we provide more discussions with other transformer-based methods; and 4) we add more experimental results to verify the effectiveness of the proposed framework and modules.

II. Related Work

A. CD of Remote Sensing Images

Technically, the task of CD takes dual-phase remote sensing images as inputs and predicts the change regions of the same places. Before deep learning, direct classification-based methods witness the great progress in CD. For example, change vector analysis (CVA) [6] is powerful in extracting pixel-level features and is widely used in CD. With the rapid improvement in image resolution, more details of objects have been recorded in remote sensing images. Therefore, many object-aware methods are proposed to improve the CD performance. For example, Tang et al. [7] propose an object-oriented CD method based on the Kolmogorov–Smirnov test. Li et al. [8] propose the object-oriented CVA to reduce the number of pseudo detection pixels. With multiple classifiers and multiscale uncertainty analysis, Tan et al. [9] build an object-based approach for complex scene CD. Although the above methods can generate CD maps from dual-phase remote sensing images, they generally deliver incomplete CD regions and irregular CD boundaries due to the limited representation ability of the extracted visual features.

With the advances of deep learning, many works improve the CD performance by extracting more discriminative features. For example, Zhang et al. [10] use a deep belief network (DBN) to extract deep features and represent the change regions by patch differences. Saha et al. [11] combine a pretrained deep CNN and traditional CVA to generate certain change regions. Hou et al. [12] take the advantages of deep features and introduce the low-rank analysis to improve the CD results. Peng and Guan [13] use saliency detection analysis and pretrained deep networks to achieve unsupervised CD. Since change regions may appear in any places, Lei et al. [14] integrate stacked denoising autoencoders (SDAEs) with multiscale superpixel segmentation to realize superpixel-based CD. Similarly, Lv et al. [15] use a stacked contractive autoencoder (SCAE) to extract temporal change features, and then adopt a clustering method to produce accurate CD maps.

Meanwhile, some methods formulate the CD task as a binary image segmentation task. Thus, CD can be finished in a supervised manner. For example, Alcantarilla et al. [16] first concatenate dual-phase images as one image with six channels. Then, the six-channel image is fed into a fully convolutional network (FCN) to realize the CD. Similarly, Peng et al. [17] combine bitemporal remote sensing images as one input for CD. Caye Daudt et al. [18] use Siamese networks to extract features for each remote sensing image, and then predict the CD maps with fused features. The experimental results prove the efficiency of Siamese networks. Furthermore, Guo et al. [19] use a fully convolutional Siamese network with a contrastive loss to measure the change regions. Zhang et al. [20] propose a deeply supervised image fusion network for CD. There are also some works focused on specific object CD. For example, Liu et al. [4] propose a dual-task constrained deep Siamese convolutional network for building CD. Jiang et al. [21] propose a pyramid feature-based attention-guided Siamese network for building CD. Lei et al. [22] propose a hierarchical paired channel fusion network for street scene CD. The aforementioned methods have shown great success in feature learning for CD. However, these methods have limited global representation capabilities and usually focus on local regions of changed objects. We find that transformers have strong characteristics in extracting global features. Thus, different from previous works, we take the advantages of transformers and propose a new learning framework for more discriminative feature representations.

B. Vision Transformers for CD

Transformers [23] are first proposed for time series tasks, such as natural language processing and speech generation. Recently, they have been applied to many computer vision tasks, such as image classification [2], [3] and person reidentification [24], [25]. Inspired by the extreme effectiveness, Zhang et al. [26] deploy a Swin Transformer structure [3] with a U-Net [27] for remote sensing image CD. Zheng et al. [28] design a deep multitask encoder–transformer–decoder (METD) architecture for semantic CD. Wang et al. [29] incorporate a Siamese vision transformer (SViT) into a feature difference framework for CD. To take the advantages of both the transformers and CNNs, Wang et al. [30] propose to combine a transformer and a CNN for remote sensing image CD. Li et al. [31] propose an encoding–decoding hybrid framework for CD, which has the advantages of both the transformers and U-Net. Bandara and Patel [32] unify hierarchically structured transformer encoders with multilayer perception (MLP) decoders in a Siamese network to efficiently render multiscale long-range details for accurate CD.

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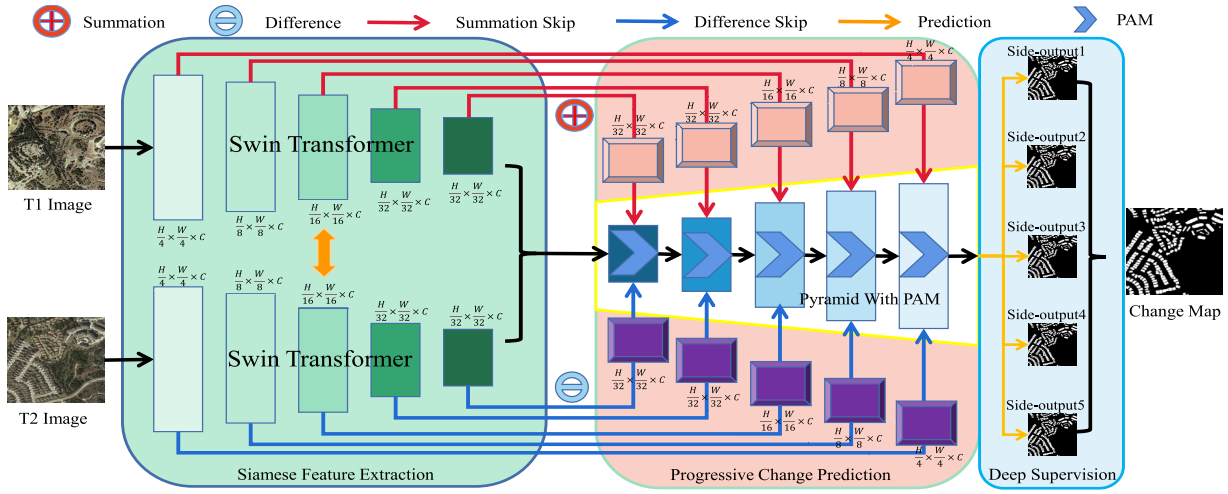


Fig. 1. Overall structure of our proposed framework (TransY-Net). First, a typical SFE network with shared Swin Transformers is used to extract multilevel feature maps from dual-phase remote sensing images. Then, the DFE is introduced to highlight the change regions with summation features and difference features. Afterward, the PCP is adopted to encode and integrate multilevel features progressively for the final change map prediction. To improve the representation ability, a pyramid structure with a PAM is used with additional interdependencies through spatial and channel attentions. Finally, the DS is used with multiple boundary-aware loss functions to train the whole framework.

Chen et al. [33] propose a bitemporal image transformer (BIT) to efficiently and effectively model contexts within the spatial–temporal domain for CD. Ke and Zhang [34] propose a hybrid transformer with token aggregation for remote sensing image CD. Song et al. [35] combine the multiscale Swin Transformer and a deeply supervised network for CD. All these methods have shown that transformers can model the interpatch relationships for strong feature representations. However, these CD methods do not take the full abilities of transformers in multilevel feature learning. Different from the existing transformer-based CD methods, our proposed approach handles incomplete CD regions and irregular CD boundaries. Besides, we use a Siamese structure to process dual-phase remote sensing images and introduce a pyramid structure to aggregate multilevel features from transformers for feature enhancement.

C. Feature Pyramid (FP) Methods in Remote Sensing

Multiscale features play an important role in remote sensing image processing, including CD. As a typical multiscale feature fusion method, FP network (FPN) [36] is first proposed for object detection from natural images, and it can aggregate multiscale features in a coarse-to-fine manner. Recently, it also shows great successes in remote sensing tasks. For example, Yang et al. [37] propose a multiscale rotation dense FPN for automatic ship detection. Li et al. [38] refine the FPN and multilayer attention network for arbitrary-oriented object detection of remote sensing images. Shamsolmoali et al. [39] use a multipatch FPN for weakly supervised object detection in optical remote sensing images. Wang et al. [40] enhance the FPN with deep semantic embedding for remote sensing scene classification. Gao et al. [41] combine multiple FPNs for end-to-end road extraction. Zhang et al. [42] introduce a Laplacian FPN for small object detection. Chen et al. [43] combine an FPN and pixel pair matching for water-body segmentation. We note that our work is indeed inspired by the classical FPN. However, our work introduces advanced transformers into the FPN, which can capture more long-range contextual information. Besides, our pyramid structure can improve the feature representation ability with additional interdependencies through spatial and channel attentions. They are different from the classical FPN.

III. Proposed Approach

As shown in Fig. 1, our proposed framework (TransY-Net) has a Y-shape structure, and includes four key components, i.e., Siamese feature extraction (SFE), deep feature enhancement (DFE), progressive change prediction (PCP), and deep supervision (DS). By taking dual-phase images as inputs, SFE first extracts multilevel visual features through two weight-shared Swin Transformers. Then, DFE uses the multilevel visual features to generate summation features and difference features, which highlight the change regions with temporal information. Afterward, by integrating all above features, PCP introduces a pyramid structure grafted with a PAM for the final CD prediction. Finally, to train our proposed framework, DS is introduced to achieve the deeply supervised learning with multiple boundary-aware loss functions for each feature level. We will elaborate on these key modules in Sections III-A–III-D.

A. Siamese Feature Extraction

Following previous works, we introduce a Siamese structure to extract multilevel features from the dual-phase remote sensing images. More specifically, the Siamese structure contains two encoder branches, which share learnable weights and are used for the multilevel feature extraction of remote sensing images at temporal phase 1 (T1) and temporal phase 2 (T2), respectively. As shown in the left part of Fig. 1, we take the Swin Transformer [3] as the basic backbone of the

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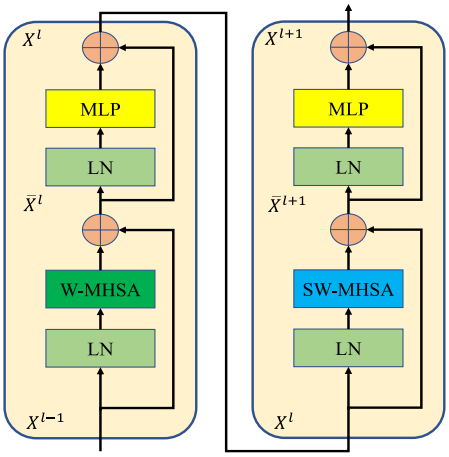


Fig. 2. Basic structure of the used Swin Transformer block.

Siamese structure, which involves five stages in total. Different from other typical transformers [2], [23], the Swin Transformer replaces the standard multihead self-attention (MHSA) with window-based MHSA (W-MHSA) and shifted W-MHSA (SW-MHSA), to reduce the computational complexity of the global self-attention. To improve the representation ability, the Swin Transformer also introduces the MLP, LayerNorm (LN) layers, and residual connections. Fig. 2 shows the basic structure of the Swin Transformer block used in this work. Technically, the calculation formulas of all the procedures are given as follows:

$$\bar{\mathbf{X}}^l = \text{W-MHSA}(\text{LN}(\mathbf{X}^{l-1})) + \mathbf{X}^{l-1} \tag{1}$$

$$\mathbf{X}^l = \text{MLP}(\text{LN}(\bar{\mathbf{X}}^l)) + \bar{\mathbf{X}}^l \tag{2}$$

$$\bar{\mathbf{X}}^{l+1} = \text{SW-MHSA}(\text{LN}(\mathbf{X}^l)) + \mathbf{X}^l \tag{3}$$

$$\mathbf{X}^{l+1} = \text{MLP}(\text{LN}(\bar{\mathbf{X}}^{l+1})) + \bar{\mathbf{X}}^{l+1} \tag{4}$$

where $\bar{\mathbf{X}}$ is the output with the W-MHSA or SW-MHSA module, and $\mathbf{X}$ is the output with the MLP module. At each stage of the original Swin Transformers, the feature resolution is halved, while the channel dimension is doubled. More specifically, the feature resolution is reduced from $(H/4) \times (W/4)$ to $(H/32) \times (W/32)$, and the channel dimension is increased from C to $8C$. To take advantages of global-level information, we introduce an additional Swin Transformer block to enlarge the receptive field of the feature maps. Besides, to reduce the computation, we uniformly change the channel dimension to C and generate encoded features $[\mathbf{E}_{T1}^1, \mathbf{E}_{T1}^2, \ldots, \mathbf{E}_{T1}^5]$, and $[\mathbf{E}_{T2}^1, \mathbf{E}_{T2}^2, \ldots, \mathbf{E}_{T2}^5]$ for the T1 and T2 images, respectively. Based on the weight-shared Swin Transformers, the multilevel features can be extracted. In general, features in the high-level capture global semantic information, while features in the low-level retain local detail information. Both of them help the detection of change regions.

B. Deep Feature Enhancement

In complex scenarios, there are many visual challenges for remote sensing image CD. Thus, only depending on the above features is not enough. To highlight the change regions, we propose to enhance the multilevel visual features with feature summation and difference, as shown in the top part and bottom part of Fig. 1. Here, we note that the difference operation is a typical method for highlighting the changed regions. While the summation operation is also a useful method for feature fusion. When using the summation of two-stream features, the common information is enhanced. It is very useful for the CD as verified in [44]. More specifically, we first perform a pointwise feature summation and difference, and then introduce a contrast feature associated with each local feature. The enhanced features can be represented as

$$\bar{\mathbf{E}}_S^k = \text{ReLU}\big(\text{BN}\big(\text{Conv}\big(\mathbf{E}_{T1}^k + \mathbf{E}_{T2}^k\big)\big)\big) \tag{5}$$

$$\bar{\mathbf{E}}_{\text{SC}}^{k,m} = \bar{\mathbf{E}}_S^k - \text{Pool}^m\big(\bar{\mathbf{E}}_S^k\big) \tag{6}$$

$$\mathbf{E}_S^k = \left[\bar{\mathbf{E}}_S^k, \bar{\mathbf{E}}_{\text{SC}}^{k,3}, \ldots, \bar{\mathbf{E}}_{\text{SC}}^{k,9}\right] \tag{7}$$

$$\bar{\mathbf{E}}_D^k = \text{ReLU}\big(\text{BN}\big(\text{Conv}\big(\mathbf{E}_{T1}^k - \mathbf{E}_{T2}^k\big)\big)\big) \tag{8}$$

$$\bar{\mathbf{E}}_{\text{dc}}^{k,m} = \bar{\mathbf{E}}_D^k - \text{Pool}^m\big(\bar{\mathbf{E}}_D^k\big) \tag{9}$$

$$\mathbf{E}_D^k = \left[\bar{\mathbf{E}}_D^k, \bar{\mathbf{E}}_{\text{dc}}^{k,3}, \ldots, \bar{\mathbf{E}}_{\text{dc}}^{k,9}\right] \tag{10}$$

where $\mathbf{E}_S^k$ and $\mathbf{E}_D^k$ $(k = 1, 2, \ldots, 5)$ are the enhanced features with pointwise summation and difference, respectively. ReLU is the rectified linear unit, BN is the batch normalization, Conv is a 1×1 convolution, and Pool^m is a $m \times m$ average pooling with appropriate paddings $(m \in \{3, 5, 7, 9\})$. [,] is the feature concatenation in the channel. In fact, $\bar{\mathbf{E}}_{\text{SC}}^{k,m}$ and $\bar{\mathbf{E}}_{\text{dc}}^{k,m}$ capture contrast features and can make change regions stand out from their surrounding background. Meanwhile, in most cases, keeping the original features shows better results due to the rich contextual information. Thus, we concatenate them with contrast features. Through the proposed DFE, more information of change regions and boundaries are highlighted with temporal information. Thus, the framework can make the extracted features more discriminative and obtain better CD results. We refer the readers to [45] for more insights.

C. Progressive Change Prediction

Since change regions can be any shapes and appear at any scales, we should consider the CD predictions in various cases. Inspired by the FP [36], we propose a PCP method, as shown in the middle part of Fig. 1. To improve the representation ability, a pyramid structure with a PAM is used with additional interdependencies through spatial and channel attentions. The structure of the proposed PAM is illustrated in Fig. 3. The PAM first takes the summation features and difference features as inputs, and then a spatial-level attention (SA) and a channel-level attention (CA) are jointly applied to enhance the features related to change regions. In addition, as we all know, residual connections [1] are famous, popular, and efficient structures in current deep models. It alleviates the vanishing gradient problem and accelerates the training convergence. Thus, we further introduce a residual connection to improve the learning ability. The final feature map can be obtained by a 1×1 convolution. Formally, the PAM can be represented as

$$\mathbf{F}^k = \text{ReLU}\big(\text{BN}\big(\text{Conv}\big(\big[\mathbf{E}_S^k, \mathbf{E}_D^k\big]\big)\big)\big) \tag{11}$$

$$\mathbf{F}_{\text{SA}}^k = \mathbf{F}^k * \sigma\big(\text{Conv}(\text{SAC}(\mathbf{F}^k))\big) \tag{12}$$

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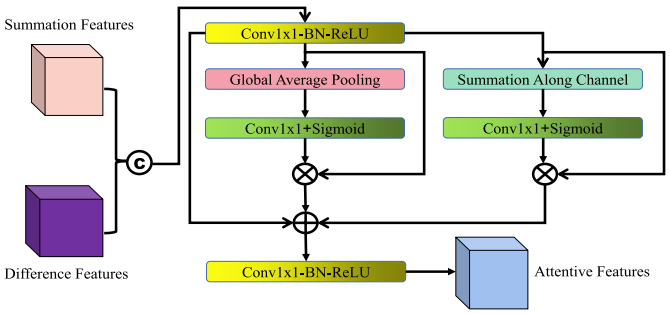


Fig. 3. Structure of our proposed PAM.

$$\mathbf{F}_{\mathrm{CA}}^{k} = \mathbf{F}^{k} * \sigma(\mathrm{Conv}(\mathrm{GAP}(\mathbf{F}^{k}))) \tag{13}$$

$$\mathbf{F}_{A}^{k} = \mathrm{Conv}\left(\mathbf{F}_{\mathrm{SA}}^{k} + \mathbf{F}_{\mathrm{CA}}^{k} + \mathbf{F}^{k}\right) \tag{14}$$

where σ is the Sigmoid function, SAC is the summation along the channel, and GAP is the global average pooling.

To achieve the PCP, we build the decoder pyramid grafted with a PAM as follows:

$$\mathbf{F}_{P}^{k} = \begin{cases} \mathbf{F}_{A}^{k}, & k = 5 \\ \mathrm{UM}\left(\mathrm{SwinBlock}^{n}\left(\mathbf{F}_{P}^{k+1}\right)\right) + \mathbf{F}_{A}^{k}, & 1 \le k < 5 \end{cases} \tag{15}$$

where UM is the patch unmerging block for feature upsampling [3], and SwinBlock^n is the Swin Transformer block with n layers. From the above formula, one can see that our PCP can make full use of the interdependencies within spatial and channel and progressively aggregate multilevel features to improve the perception ability of the change regions. Here, we note that the pyramid structure is complementary to residual connections. Many previous works have already verified this fact. Thus, it is reasonable to introduce residual connections under the premise of a pyramid structure.

D. Loss Function

To optimize our framework, DS is used to achieve the deeply supervised learning [46], [47], [48] with multiple boundary-aware loss functions for each feature level. The overall loss is defined as the summation loss over all the side outputs and the final fusion prediction. Specifically, we first take the features of the PCP, i.e., $\mathbf{F}_{P}^{k}(k = 1, 2, \ldots, 5)$, and use a deconvolutional layer for the prediction $\mathbf{P}^{s}$ as side outputs. Then, we concatenate them for the final fusion prediction

$$\mathbf{P}^{f} = \mathrm{Conv}[\mathbf{P}^{1}, \ldots, \mathbf{P}^{S}]. \tag{16}$$

All the side outputs and the final fusion prediction are supervised by the proposed hybrid loss

$$\mathcal{L} = \mathcal{L}^{f} + \sum_{s=1}^{S} \alpha_{s} \mathcal{L}^{s} \tag{17}$$

where $\mathcal{L}^{f}$ is the loss of the final fusion prediction, and $\mathcal{L}^{s}$ is the loss of the sth side-output, respectively. S denotes the total number of the side outputs, and α_s is the weight for each level loss. Our method includes five side outputs, i.e., $S = 5$.

To obtain complete CD regions and regular CD boundaries, we define $\mathcal{L}^{f}$ or $\mathcal{L}^{s}$ as a combined loss with three terms

$$\mathcal{L}^{f/s} = \mathcal{L}_{\mathrm{WBCE}} + \mathcal{L}_{\mathrm{SSIM}} + \mathcal{L}_{\mathrm{SIoU}} \tag{18}$$

where $\mathcal{L}_{\mathrm{WBCE}}$ is the weighted binary cross-entropy loss, $\mathcal{L}_{\mathrm{SSIM}}$ is the structural similarity loss, and $\mathcal{L}_{\mathrm{SIoU}}$ is the soft intersection over union (IoU) loss. $\mathcal{L}_{\mathrm{WBCE}}$ provides a probabilistic measure of similarity between the prediction and ground truth from a pixel-level view. $\mathcal{L}_{\mathrm{SSIM}}$ captures the structural information of change regions in patch level. $\mathcal{L}_{\mathrm{SIoU}}$ is inspired by measuring the similarity of two sets and yields a global similarity in CD map level. More specifically, given the ground-truth probability $g_l(\mathbf{x})$ and the estimated probability $p_l(\mathbf{x})$ at pixel $\mathbf{x}$ belong to the class l, the $\mathcal{L}_{\mathrm{WBCE}}$ loss function is defined as

$$\mathcal{L}_{\mathrm{WBCE}} = -\sum_{\mathbf{x}} w(\mathbf{x}) g_l(\mathbf{x}) \log(p_l(\mathbf{x})). \tag{19}$$

Here, we use weights $w(\mathbf{x})$ to adapt the loss function to the challenges that we have encountered in CD: the class imbalance and the errors along CD boundaries. Given the frequency f_l of class l in the training data, the indicator function I, the training prediction T, and the gradient operator ∇, then the weights are defined as

$$w(\mathbf{x}) = \sum_{l} I(T(\mathbf{x} == l)) \frac{\mathrm{median}(\mathbf{f})}{f_l} + w_0 I(|\nabla T(\mathbf{x})| > 0) \tag{20}$$

where $\mathbf{f} = [f_1, \ldots, f_L]$ is the vector of all the class frequencies. The first term models the median frequency balancing [49] and compensates for the class imbalance problem by highlighting classes with a low probability. The second term puts higher weights on the CD boundaries to emphasize on the correct detection of contours.

$\mathcal{L}_{\mathrm{SSIM}}$ loss considers a local neighborhood of each pixel [50]. Let $\hat{\mathbf{x}} = \{\mathbf{x}_j : j = 1, \ldots, N^2\}$ and $\hat{\mathbf{y}} = \{\mathbf{y}_j : j = 1, \ldots, N^2\}$ be the pixel values of two corresponding patches (size: $N \times N$) cropped from the prediction P and the ground truth G, respectively, $\mathcal{L}_{\mathrm{SSIM}}$ loss is defined as

$$\mathcal{L}_{\mathrm{SSIM}} = 1 - \frac{(2\mu_{\mathbf{x}}\mu_{\mathbf{x}} + \epsilon)(2\sigma_{\mathbf{xy}} + \epsilon)}{(\mu_{\mathbf{x}}^2 + \mu_{\mathbf{y}}^2 + \epsilon)(\sigma_{\mathbf{x}}^2 + \sigma_{\mathbf{y}}^2 + \epsilon)} \tag{21}$$

where $\mu_{\mathbf{x}}$, $\mu_{\mathbf{y}}$ and $\sigma_{\mathbf{x}}$, $\sigma_{\mathbf{y}}$ are the mean and standard deviations of $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$, respectively. $\sigma_{\mathbf{xy}}$ is their covariance. Here, $\epsilon = 10^{-4}$ is used to avoid dividing by zero.

In this work, one metric of interest at test time is the IoU. Thus, we also introduce the soft IoU loss [51], which is differentiable for model learning. $\mathcal{L}_{\mathrm{SIoU}}$ is defined as

$$\mathcal{L}_{\mathrm{SIoU}} = 1 - \frac{\sum_{\mathbf{x}} p_l(\mathbf{x}) g_l(\mathbf{x})}{\sum_{\mathbf{x}} [p_l(\mathbf{x}) + g_l(\mathbf{x}) - p_l(\mathbf{x}) g_l(\mathbf{x})]}. \tag{22}$$

When using all the above losses, $\mathcal{L}_{\mathrm{WBCE}}$ loss can relieve the class imbalance problem for change pixels, $\mathcal{L}_{\mathrm{SSIM}}$ loss highlights the local structure of change boundaries, and $\mathcal{L}_{\mathrm{SIoU}}$ loss gives more focus on the overall change regions. Thus, we can obtain better CD results and make the framework easier to optimize.

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IV. Experiments

In this section, we perform extensive experiments to verify the effectiveness of the proposed framework. We first introduce the used datasets, evaluation metrics, and implementation details. Then, we compare the proposed method with other outstanding CD methods. Finally, we perform ablation studies to verify the effectiveness of key modules with quantitative and qualitative comparisons.

A. Datasets

LEVIR-CD [52] is a public large-scale remote sensing CD dataset. It contains 637 image pairs with a 1024 × 1024 resolution (0.5 m). We follow its default dataset split and crop the original images into small patches of size 256 × 256 with no overlapping. Therefore, we obtain 7120/1024/2048 pairs of image patches for training/validation/test, respectively.

WHU-CD [53] is a public building CD dataset. It contains one pair of high-resolution (0.075 m) aerial images of size 32 507 × 15 354. As no predefinite data split is widely used, we crop the original image into small patches of size 256 × 256 with no overlap and randomly split them into three parts: 6096/762/762 for training/validation/test, respectively.

SYSU-CD [54] is also a public building CD dataset. It contains 20 000 pairs of high-resolution (0.5 m) images of size 256 × 256. We follow its default dataset split for experiments. There are 12 000/4000/4000 pairs of image patches for training/validation/test, respectively.

Google-CD [55] is a very recent and public CD dataset. It contains 19 image pairs, originating from Google Earth Map. The image resolutions range from 1006 × 1168 pixels to 4936 × 5224 pixels. As WHU-CD, we also crop the original images into small patches of size 256 × 256 with no overlap and randomly split them into three parts: 2504/313/313 for training/validation/test, respectively.

In addition, we adopt two SAR image CD datasets [56] to verify the generalization of our method. These two datasets were gathered from disaster-stricken environments, which are related to flood and ice breakup situations, respectively. The Ottawa dataset is captured by the RADARSAT SAR sensor in May and August 1997 and changes are caused by floods. The size of each image is 290 × 350 pixels. The Sulzberger dataset is a part of Sulzberger Ice Shelf, which is provided by the European Space Agency's ENVISAT satellite. The size of the image is 256 × 256 pixels. It clearly shows the breakup of an ice shelf caused by a tsunami in March 2011.

B. Evaluation Metrics

To verify the performance of our framework and other compared methods, we follow previous works [30], [32], [33], [34] and use F1 and IoU scores with regard to the change class as the primary evaluation metrics. In addition, we also report the precision and recall of the change category, overall accuracy (OA), and receiver operating characteristic (ROC) curve. To evaluate the performance of regional boundaries, we follow previous works [57], [58] and adopt the mean boundary accuracy (mBA) as the metric.

C. Implementation Details

We perform experiments with the PyTorch toolbox and one NVIDIA A30 GPU. We use the mini-batch SGD algorithm to train our framework with an initial learning rate 10^{-3}, moment 0.9, and weight decay 0.0005. The batch size is set to 6. For the SFE backbone, we adopt the Swin Transformer pretrained on the ImageNet-22k classification task [60]. To fit the input size of the pretrained Swin Transformer, we uniformly resize image patches to 384 × 384. For other layers, we randomly initialize them and set the learning rate with 10 times than the initial learning rate. We train the framework with 100 epochs. The learning rate decreases to 1/10 of the initial learning rate at every 20th epoch. To improve the robustness, data augmentation is performed by random rotation and flipping of the input images. For the loss function in model training, the weight parameters of each level are set equally. We release the source code at https://github.com/Drchip61/TransYNet.

D. Comparisons With State-of-the-Arts

In this section, we compare the proposed method with other outstanding methods four optical and two SAR image CD datasets. These experimental results fully verify the effectiveness of our proposed framework and modules.

1) Quantitative Comparisons: We present the comparative results in Tables I and II. The results clearly show that our method delivers excellent performance. More specifically, our method achieves the F1 and IoU scores of 91.90% and 83.64% on the LEVIR-CD dataset, respectively. They are much better than most of the previous methods. Besides, compared with other transformer-based methods, such as BIT [33], H-TransCD [34] and ChangeFormer [32], our method shows consistent improvements in terms of all the evaluation metrics. When compared with our previous method FTN [5], the method in this article can achieve better results in almost all the metrics. On the WHU-CD dataset, our method shows significant improvements with the F1 and IoU scores of 93.38% and 87.58%, respectively. In comparison to the second-best method (FTN), our method improves the F1 and IoU scores by 1.2% and 2.1%, respectively. On the SYSU-CD dataset, our method achieves the F1 and IoU scores of 82.84% and 70.71%, respectively. The SYSU-CD dataset includes more large-scale change regions. We believe that the improvements are mainly based on the proposed DFE. On the Google-CD dataset, our method shows much better results than other compared methods. In fact, our method achieves the F1 and IoU scores of 86.04% and 75.50%, respectively. We note that the Google-CD dataset is recently proposed and it is much challenging than other three datasets. We also note that the performance of precision, recall, and OA is not consistent in all the methods. Our method generally achieves better recall values than most compared methods. The main reason may be that our method gives higher confidences to the change regions. To better illustrate the performance, we also present the ROC curves of some typical methods in Fig. 4. It is observed that our method achieves better results than other typical methods on the WHU-CD dataset.

2) Qualitative Comparisons: To illustrate the visual effect, we first display some typical CD results on the four optical

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TABLE I

QUANTITATIVE COMPARISONS ON THE LEVIR-CD AND WHU-CD DATASETS. THE BEST AND THE SECOND BEST ARE IN BOLD AND UNDERLINE, RESPECTIVELY. — MEANS THE RESULTS OF THE CORRESPONDING METHODS ARE MISSING

Methods	LEVIR-CD					WHU-CD				
	Pre.	Rec.	F1	IoU	OA	Pre.	Rec.	F1	IoU	OA
FC-EF [18]	86.91	80.17	83.40	71.53	98.39	71.63	67.25	69.37	53.11	97.61
FC-Siam-Diff [18]	89.53	83.31	86.31	75.92	98.67	47.33	77.66	58.81	41.66	95.63
FC-Siam-Conc [18]	91.99	76.77	83.69	71.96	98.49	60.88	73.58	66.63	49.95	97.04
BiDateNet [4]	85.65	89.98	87.76	78.19	98.52	78.28	71.59	74.79	59.73	81.92
U-Net++MSOF [17]	90.33	81.82	85.86	75.24	98.41	91.96	89.40	90.66	82.92	96.98
DTCDSCN [4]	88.53	86.83	87.67	78.05	98.77	63.92	82.30	71.95	56.19	97.42
DASNet [4]	80.76	79.53	79.91	74.65	94.32	68.14	73.03	70.50	54.41	97.29
STANet [52]	83.81	**91.00**	87.26	77.40	98.66	79.37	85.50	82.32	69.95	98.52
MSTDSNet [35]	85.52	90.84	88.10	78.73	98.56	—	—	—	—	—
IFNet [20]	**94.02**	82.93	88.13	78.77	98.87	**96.91**	73.19	83.40	71.52	98.83
SNUNet [59]	89.18	87.17	88.16	78.83	98.82	85.60	81.49	83.50	71.67	98.71
BIT [33]	89.24	89.37	89.31	80.68	98.92	86.64	81.48	83.98	72.39	98.75
H-TransCD [34]	91.45	88.72	90.06	81.92	99.00	93.85	88.73	91.22	83.85	99.24
UVACD [30]	91.90	90.70	91.30	**83.98**	**99.12**	91.45	88.72	90.06	81.92	99.00
ChangeFormer [32]	92.05	88.80	90.40	82.48	99.04	91.83	88.02	89.88	81.63	99.12
FTN [5]	92.71	89.37	91.01	83.51	99.06	93.09	91.24	92.16	85.45	99.37
Ours	92.90	89.35	**91.90**	83.64	99.07	94.68	**92.12**	**93.38**	**87.58**	**99.47**

TABLE II

QUANTITATIVE COMPARISONS ON THE SYSU-CD AND GOOGLE-CD DATASETS. THE BEST AND THE SECOND BEST ARE IN BOLD AND UNDERLINE, RESPECTIVELY. — MEANS THE RESULTS OF THE CORRESPONDING METHODS ARE MISSING

Methods	SYSU-CD					Google-CD				
	Pre.	Rec.	F1	IoU	OA	Pre.	Rec.	F1	IoU	OA
FC-EF [18]	74.32	75.84	75.07	60.09	86.02	80.81	64.39	71.67	55.85	85.85
FC-Siam-Diff [18]	**89.13**	61.21	72.57	56.96	82.11	85.44	63.28	72.71	57.12	87.27
FC-Siam-Conc [18]	82.54	71.03	76.35	61.75	86.17	82.07	64.73	72.38	56.71	84.56
BiDateNet [4]	81.84	72.60	76.94	62.52	89.74	78.28	71.59	74.79	59.73	81.92
U-Net++MSOF [17]	81.36	75.39	78.26	62.14	86.39	91.21	57.60	70.61	54.57	95.21
DASNet [4]	68.14	70.01	69.14	60.65	80.14	71.01	44.85	54.98	37.91	90.87
STANet [52]	70.76	**85.33**	77.37	63.09	87.96	89.37	65.02	75.27	60.35	82.58
DSAMNet [20]	74.81	81.86	78.18	64.18	89.22	72.12	80.37	76.02	61.32	94.93
MSTDSNet [35]	79.91	80.76	80.33	67.13	90.67	—	—	—	—	—
SRCDNet [55]	75.54	81.06	78.20	64.21	89.34	83.74	71.49	77.13	62.77	83.18
BIT [33]	82.18	74.49	78.15	64.13	90.18	**92.04**	72.03	80.82	67.81	96.59
H-TransCD [34]	83.05	77.40	80.13	66.84	90.95	85.93	81.73	83.78	72.08	97.64
FTN [5]	86.86	76.82	81.53	68.82	91.79	86.99	**84.21**	85.58	74.79	97.92
Ours	89.09	77.42	**82.84**	**70.71**	**92.44**	87.98	84.18	**86.04**	**75.50**	**97.97**

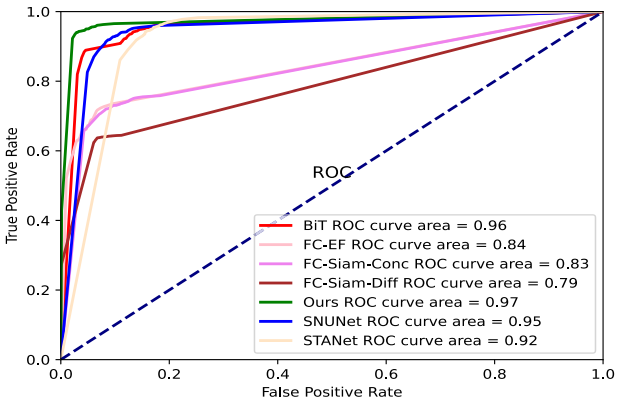


Fig. 4. ROC curves of some typical methods on the WHU-CD dataset.

image CD datasets, as shown in Figs. 5–8. From the results, one can see that our method generally shows best results. For example, when change regions have multiple scales, our method can correctly identify most of the change regions, as shown in Fig. 5. When change objects cover most of the image regions, most of current methods cannot detect them. However, our method can still detect them with clear boundaries, as shown in Fig. 6. In addition, when the change regions appear in complex scenes, our method can maintain the contour shape. While most of the compared methods fail, as shown in Fig. 7. When distractors appear in the scene, our method can reduce the effect and correctly detect the real change regions, as shown in Fig. 8. From the above visual results, we can see that our method shows superior performance than most methods.

To further verify the visual effect, we provide more hard samples and failed results in Fig. 9. As can be seen, our method performs better than most methods (first row). Most of current methods cannot detect the two small change regions in the center, while our method can accurately localize them. Besides, we also show failed examples in the second row of Fig. 9. As can be seen, all the compared methods cannot detect

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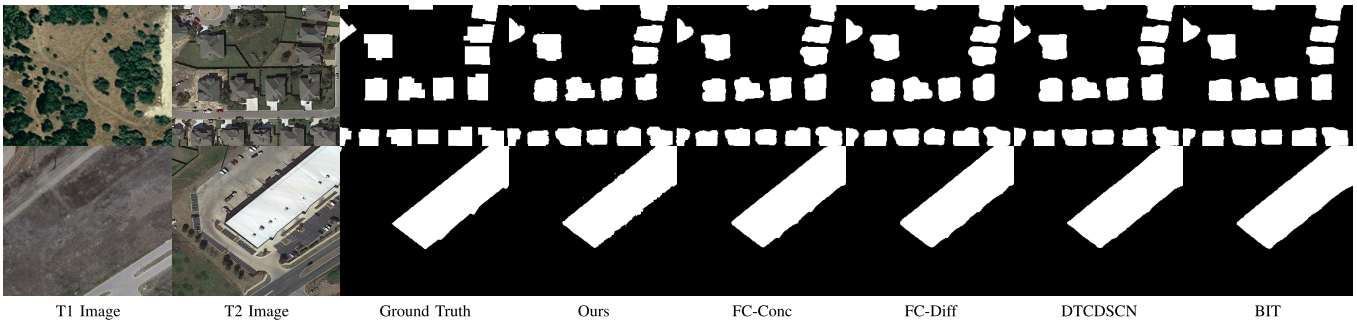


Fig. 5. Comparison of typical CD results on the LEVIR-CD dataset. Best view by zooming in.

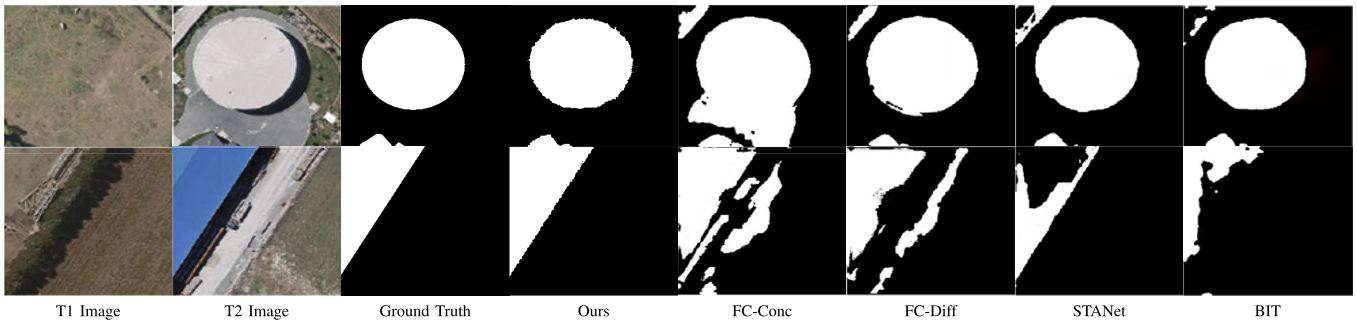


Fig. 6. Comparison of typical CD results on the WHU-CD dataset. Best view by zooming in.

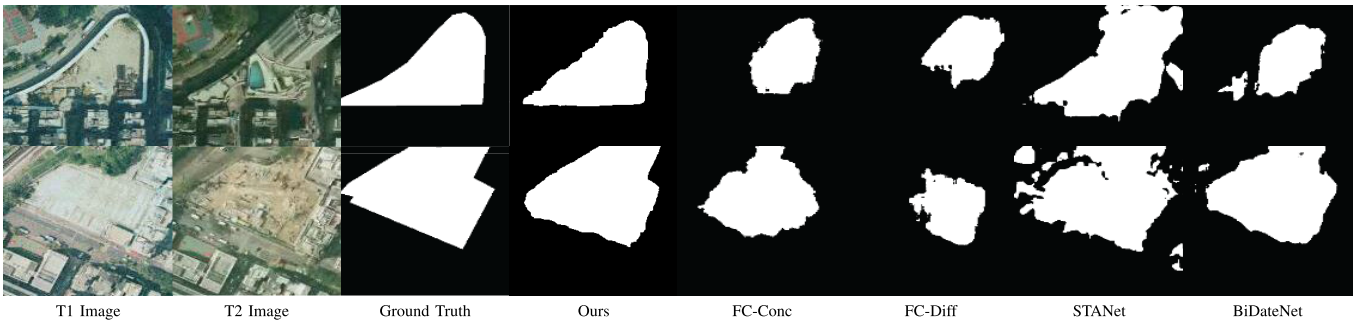


Fig. 7. Comparison of typical CD results on the SYSU-CD dataset. Best view by zooming in.

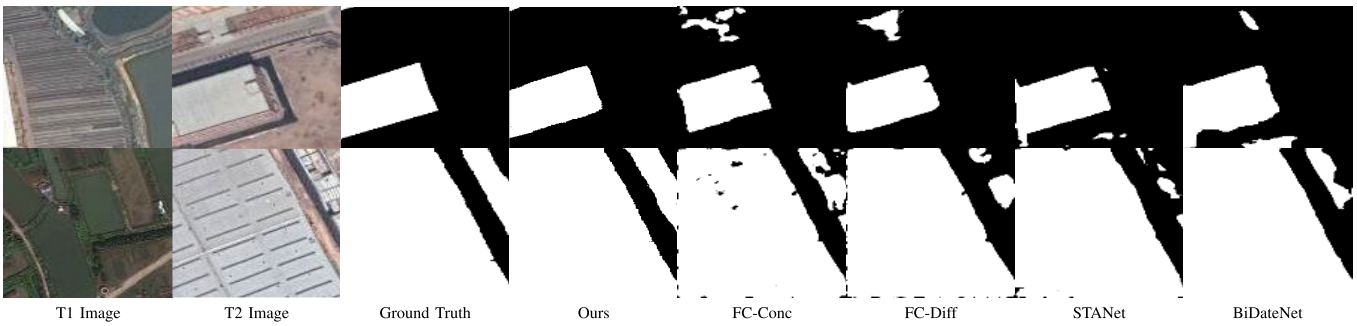


Fig. 8. Comparison of typical CD results on the Google-CD dataset. Best view by zooming in.

all the change regions. However, our method shows a much more reasonable result than other methods.

To verify the generalization of our method, Fig. 10 shows the CD results on two SAR image CD datasets. The other compared methods include PCANet [61], NR-ELM [62], DBN [63], DCNet [64], MSAPNet [65], and DDNet [66]. It can be observed that our proposed method shows better results of CD regions and boundaries. These results also demonstrate the effectiveness of our method on different types of remote sensing images and in disaster environments, including flood and tsunami.

E. More Discussions

To better clarify the model performance, we plot the dynamic results of our model during the training, validation, and testing phases (see Fig. 11). It can be observed that our

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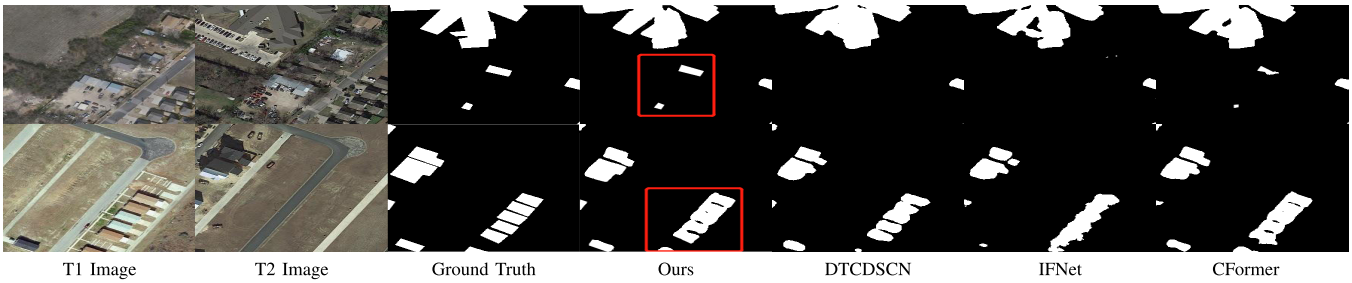


Fig. 9. Comparison of typical CD results on more hard and failed samples. Best view by zooming in.

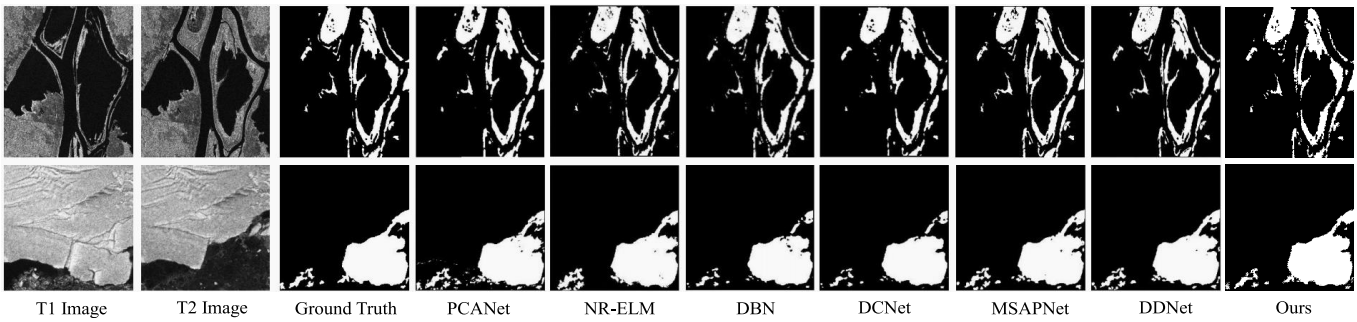


Fig. 10. Comparison of typical CD results on (top) Ottawa dataset and (bottom) Sulzberger dataset. Images are in disaster environments. Best view by zooming in.

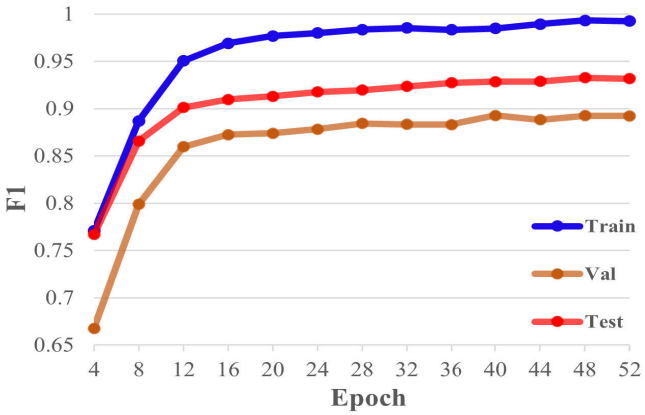


Fig. 11. Dynamic results of our model on the WHU-CD dataset.

model performs better on the training data than the test data. This is reasonable and practical since most of the current deep learning methods have similar trends.

Previous sections display the performance of changed regions. To evaluate the performance of regional boundaries, we list the boundary accuracy in Table III. It can be observed that our method shows much better results than other outstanding methods. It clearly demonstrates the effectiveness of our proposed method in improving the boundary accuracy.

We note that transformers (including fully transformer networks) are not new in current remote sensing and computer vision fields. However, there are some key differences between our work and previous methods.

1) As far as we know, our work is the earliest transformer-based one, which explicitly handles incomplete regions and irregular boundaries for remote sensing image CD.
2) In our framework, we use a Siamese structure to process dual-phase remote sensing images. Besides, we introduce a pyramid structure to aggregate multilevel visual features from transformers for feature enhancement. These designs are totally different from the existing works, especially in [67] and [68], which mainly use a simple encoder–decoder + U-Net structure for single-image feature extraction. He et al. [67] use a pyramid group transformer (PGT) as the encoder and propose an FP transformer as the decoder, which is largely based on the typical FPN structure. Meanwhile, the work in [68] simply stacks spatial pyramid transformers (SPTs) imitating the U-Net structure. Both of them are taking single images as inputs and using an encoder–decoder structure. While our framework uses a Siamese structure to process dual-phase images.
3) We use the deeply supervised learning with multiple boundary-aware loss functions. These losses are very helpful for more accurate CD. These facts make our framework more convincing in the CD techniques.

TABLE III
BOUNDARY ACCURACIES WITH DIFFERENT METHODS ON LEVIR-CD

Methods	Ours	ChangeFormer	BIT	SNUNet	STANet	FC-Diff
mBA	71.6	68.7	65.8	65.2	63.3	60.4

F. Ablation Studies

In this section, we perform extensive ablation studies to verify the effect of key components in our framework. The experiments are conducted on the LEVIR-CD dataset. However, other datasets have similar performance trends.

1) Effects of Different Siamese Backbones: As shown in rows 2 and 3 of Table IV, we introduce the VGGNet-16 [69] and Swin Transformer as the Siamese backbones. To ensure a

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Fig. 12. Visual comparisons of predicted change maps with different models. Best view by zooming in.

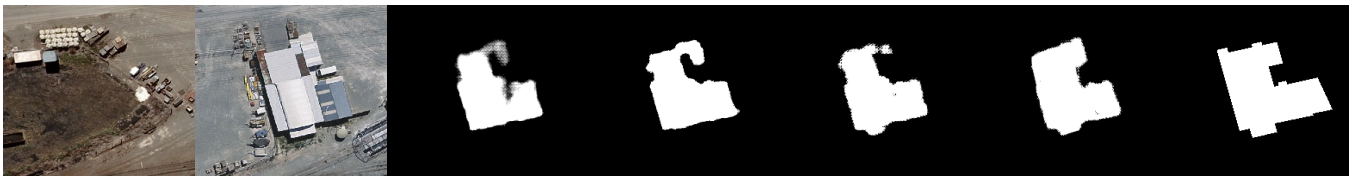


Fig. 13. Visual comparisons of predicted change maps with different losses. Best view by zooming in.

TABLE IV
PERFORMANCE COMPARISONS WITH DIFFERENT MODEL VARIANTS

Models	Pre.	Rec.	F1	IoU	OA
(a) VGGNet-16+FP	91.98	82.65	87.06	77.09	98.75
(b) SwinT+FP	91.12	87.42	89.23	80.56	98.91
(c) SwinT+DFE+FP	91.73	88.43	90.05	81.89	99.00
(d) SwinT+DFE+PCP [5]	92.71	89.37	91.01	83.51	99.06
(e) SwinT+DFE+PCP	92.90	89.35	91.90	83.64	99.07

fair comparison, we use the basic FP structure [36]. From the results, one can see that the performance with the Swin Transformer can be consistently improved in terms of recall, F1, IoU, and OA. The main reason is that the Swin Transformer has a better ability of modeling long-range dependency.

2) Effects of DFE: The fourth row of Table IV shows the effect of our proposed DFE. When compared with Model(b) Swin$T$ + FP, the DFE improves the F1 value from 89.23% to 90.05%, and the IoU value from 80.56% to 81.89%, respectively. The main reason is that our DFE considers the temporal information with feature summation and difference, which highlight the change regions.

3) Effects of PCP: To better detect multiscale change regions, we introduce the PCP, which is a pyramid structure grafted with a PAM. If we remove the PAM, the PCP will reduce a basic FP. We compare it with FP. From the results in the last two rows of Table IV, one can see that our PCP achieves a significant improvement in all the metrics. Besides, the improved PCP in this work shows better performances. Furthermore, adding the PCP also achieves a better visual effect, in which the extracted change regions are complete and the boundaries are regular, as shown in Fig. 12.

In addition, we also introduce the Swin Transformer blocks into the PCP as shown in (11). To verify the effect of different layers, we report the results in Table V. From the results, one can see that the models show better results with equal layers. The best results can be achieved with $n = 4$. With more layers, the computation is larger and the performance decreases.

4) Effects of Different Losses: In this work, we introduce multiple losses. To show the effects of these losses,

TABLE V
PERFORMANCE COMPARISONS WITH DIFFERENT DECODER LAYERS

Layers	Pre.	Rec.	F1	IoU	OA
(2,2,2,2)	91.18	87.00	89.04	80.24	98.90
(4,4,4,4)	91.65	88.42	90.01	81.83	99.00
(6,6,6,6)	91.70	88.30	89.96	81.76	98.99
(8,8,8,8)	91.55	88.47	89.98	81.79	98.99
(2,4,6,8)	92.13	85.71	88.80	79.86	98.89

TABLE VI
PERFORMANCE COMPARISONS WITH DIFFERENT LOSSES

Losses	Pre.	Rec.	F1	IoU	OA
BCE	90.68	86.91	88.75	79.78	98.88
WBCE	91.65	88.42	90.01	81.83	99.00
WBCE+SSIM	91.71	88.57	90.11	82.27	99.01
WBCE+SSIM+SIoU	92.71	89.37	91.01	83.51	99.06

we adopt the network structure in [5]. Table VI shows the results. It can be seen that using the WBCE loss can improve the F1 score from 88.75% to 90.01% and the IoU from 79.78% to 81.83%. Using the SSIM loss achieves the F1 score of 90.11% and the IoU of 82.27%. Using the SIoU loss achieves the F1 score of 91.01% and the IoU of 83.51%.

We also display some typical examples for the visual effects, as shown in Fig. 13. From the results, one can see that using the WBCE loss can help the model focus on the most change regions. With the SSIM loss, the framework can improve the structural information of the change regions. Using the SIoU loss can ensure the global completeness. As a result, combining all of them can achieve the best results, which proves the effectiveness of all the loss terms. This fact is consistent with the quantitative results in Table VI.

5) Scaling to Higher Resolutions: Remote sensing images always hold large resolutions. The resolution concern is very valuable. In fact, our work can process a higher resolution with SwinT-Base/Small/Tiny. However, we adopt a low resolution (256×256), mainly considering the fairness. Most of the compared methods use cropping for generating input images.

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TABLE VII
PERFORMANCES WITH DIFFERENT INPUT RESOLUTIONS ON LEVIR-CD

Resolution	Pre.	Rec.	F1	IoU	OA	Flops(G)
256×256	92.90	89.35	91.90	83.64	99.07	48
384×384	93.01	90.11	92.12	84.20	99.10	151
512×512	93.23	90.35	92.25	85.10	99.32	201

Thus, we follow them and realize fair comparisons. Table VII shows the performance analysis with different resolutions on LEVIR-CD. One can see that our method can naturally scale to higher resolutions and show slightly better results.

V. CONCLUSION AND FUTURE WORK

In this work, we propose a new learning framework named TranY-Net for CD of dual-phase remote sensing images. It improves the feature extraction from a global view and combines multilevel visual features in a pyramid manner. Technically, we first use a Siamese network with the pretrained Swin Transformers to extract long-range dependency information. Then, we introduce a pyramid structure to aggregate multilevel visual features, improving the feature representation ability. Finally, we use the deeply supervised learning with multiple loss functions for model training. Extensive experiments on four public CD benchmarks demonstrate that our proposed framework shows better performances than most state-of-the-art methods. However, our methods have some shortcomings, such as high computation and the need of image dense labeling. In future works, we will explore more efficient structures of transformers to reduce the computation. We will also develop unsupervised or weakly supervised methods to relieve the burden of remote sensing image labeling. In addition, since our method takes dual-phase images, it can be easily used for other similar multimodal/temporal tasks, such as RGB-D/T image segmentation, MRI-CT image fusion, and video segmentation. We will verify them in the computer vision field.

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